VISHVANATH S

Madurai, Tamil Nadu 625012

9944135600 | vishvanath18112005@gmail.com | linkedin.com/in/vishvanath07/ | github.com/in/vishvanath07

Training

Semiconductor Summer Trainee Jun ‘25-Jun’25

CeNSE, IISc Bangalore

- Gained comprehensive exposure to the end-to-end semiconductor manufacturing flow, spanning from material science to chip packaging and silicon wafer tapeout.
- Analyzed foundry workflows including the transition from RTL to GDSII, ensuring design readiness for physical fabrication.
- Visualized Silvaco TCAD for device-level modeling and process simulation to analyze semiconductor material properties and transistor characteristics.
- Analysed physical layout design and verification using Cadence Virtuoso, focusing on standard cell architectures and interconnect routing

Design and Verification Trainee Jun ‘26-Jul ‘26

Futurewiz

- Completed hands-on training in digital design and functional verification, covering the RTL design flow from specification through simulation and debug.
- Designed combinational and sequential digital circuits, including finite state machines, using **Verilog** HDL with synthesizable RTL coding practices.
- Built layered verification testbenches in **SystemVerilog** using interfaces, classes, randomization, and constraints for stimulus generation.
- Applied SystemVerilog assertions, functional coverage, and scoreboard-based checking to validate design behavior and measure verification completeness.

Projects

APB Slave Verification Environment – SystemVerilog UVM-Style Testbench Jun ’26

- Designed and implemented a complete SystemVerilog-based verification environment for an AMBA APB (Advanced Peripheral Bus) slave, following a layered testbench architecture to validate protocol compliance and functional correctness.
- Developed a modular testbench comprising interface, transaction, generator, driver, monitor, scoreboard, and environment components, enabling reusable and scalable stimulus generation and response checking.
- Implemented a constrained-random stimulus generation mechanism with an error-learning feedback loop, dynamically adjusting constraints to improve coverage of corner-case and error scenarios across test runs.
- Built a scoreboard to perform automated response checking against expected APB read/write behavior, flagging protocol violations and functional mismatches in real time.
- Authored the APB slave DUT (Design Under Test) independently and verified it against the custom testbench, producing a structured verification report summarizing functional coverage and test outcomes.

  Tech Stack: SystemVerilog, APB Protocol, Functional Verification, Constrained-Random Verification, Testbench Architecture, Scoreboard-Based Checking

RRAM-Based In-Memory Computing Spiking Neural Network for Edge AI Feb ’26

- Engineered a neuromorphic computing framework utilizing Resistive RAM (RRAM) crossbar arrays to enable energy-efficient in-memory computation for edge AI inference.
- Designed and analyzed 1T1R RRAM cell architecture and implemented crossbar-based synaptic weight storage to perform analog matrix–vector multiplication directly within memory.
- Developed a Spiking Neural Network (SNN) neuron model based on spike-response dynamics to support event-driven computation with ultra-low power consumption.
- Performed Monte Carlo variability analysis to evaluate device-level effects such as threshold variation, conductance fluctuation, and switching instability on spike generation reliability.
- Simulated I–V hysteresis characteristics of RRAM devices and validated resistive switching behavior for synaptic weight modulation.

  Tech Stack: Cadence Virtuoso, Verilog/Verilog-A, RRAM Device Modeling, Spiking Neural Networks (SNN), Monte Carlo Analysis

Self-Healing Computing System – Bharat Electronics Limited Sep'25

- Designed and implemented a 4-bit Arithmetic Logic Unit (ALU) with integrated fault detection and Built-In Self-Test (BIST) mechanism using a golden-signature comparator to enable autonomous fault identification in digital hardware systems.
- Implemented a reference (golden) output model alongside the ALU datapath, with a dedicated comparator block flagging real-time mismatches between actual and expected results to detect functional faults during operation.
- Designed a self-healing recovery mechanism using partial reconfiguration on a Xilinx FPGA platform, where a detected fault triggers dynamic isolation of the faulty logic block and live replacement with a redundant module — healing the system in real time without halting or interrupting ongoing operation.
- Validated fault-detection coverage through manually crafted fault-injection test vectors across multiple input scenarios, verifying comparator response and correct BIST-to-healing trigger behavior.
- Performed functional verification and hardware testing using simulation and on-board deployment to confirm correct ALU operation and successful fault recovery

  Tech Stack: Verilog, Xilinx Vivado, FPGA (Nexys4), BIST/Golden-Signature Comparison, Partial Reconfiguration, Fault Injection Testing, Digital System Design, Fault-Tolerant Computing

Certificates

FPGA Programming and Architecture Oct ‘25
NIELIT Calicut, NPTEL

Database Management System Oct ‘25
IIT Kharagpur, NPTEL

RTL to GDS II Nov‘25
Digital Badge, Cadence Design System

Skills

Languages : C++, Verilog, System verilog.
FPGA : NEXY 4 DDR2 (Artix-7), Spartan 6.
VLSI Skills : RTL Design, Verification, Testing, Fabrication.
Hardware & Design : Digital Circuits, Analog Circuits, Circuit Design, Neuromorphic Computing, Memory Design, Built in Self test (BIST).
Simulation & EDA Tools : Cadence Spectre, Cadence Virtuoso, Cadence NC-Sim, Xilinx Vivado.
Soft Skills : Problem-Solving, Time Management, Team Collaboration, Adaptability.

Achievements

Secured First position Mar ’25
Fit-Fusion, Cognizance, IIT Roorkee - 2025

First Runner up Feb ’25
R2VC workshop, IIT Roorkee - 2025

Education

Lovely Professional University, Punjab 2023 – 2027
Electronics and communication Engineering — CGPA: 7.81 Jalandhar, Punjab

Sri Aurobindo Mira Universal, school 2021 – 2023
12th with Science. Percentage-70.8 Madurai, Tamil Nadu

Sri Aurobindo Mira Universal, school 2020 – 2021
10th with Science. Percentage-80.8 Madurai, Tamil Nadu